THE DIMENSION OF THE LOCAL RING OF THE FIBRE IS ETALE LOCAL ON THE SOURCE-AND-TARGET

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The definition on which the statement depends is delivered as a context span (role=context) rather than as a navigation stub, so the reader sees the notion of localness at a point being tested.

Editorial note. The counted spans carry 2 navigation stubs; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- Definition of etale localness on the source-and-target for morphisms of germs. Same chapter, Definition definition-local-source-target-at-point, invoked at proof line 8232 and delivered as a context span. Supplies the shape of the diagram that must be checked.
- Existence of an etale morphism of fibres over the relevant points. Same chapter, Lemma lemma-etale-on-fiber, invoked at proof lines 8233-8235. Supplies the morphism of fibres used for the comparison.
- Invariance of the dimension of the local ring under an etale morphism of fibres. Same chapter, Lemma lemma-dimension-local-ring-local, invoked at proof lines 8236-8237. Supplies the equality of the two local ring dimensions.

Source reference. Original lemma 04N8: sources/stacks-descent/pinned-original-source.tex, source lines 5454–5454. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 04NI: sources/stacks-descent/pinned-original-source.tex, source lines 8186–8186. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:descent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Definition 0.1. Let $\mathcal{Q}$ be a property of morphisms of germs of schemes. We say $\mathcal{Q}$ is *étale local on the source-and-target* if for any commutative diagram

$$\begin{array}{ccc} (U', u') & \xrightarrow[h']{} & (V', v') \\ {\scriptstyle a}\downarrow & & \downarrow{\scriptstyle b} \\ (U, u) & \xrightarrow{h} & (V, v) \end{array}$$

of germs with étale vertical arrows we have $\mathcal{Q}(h) \Leftrightarrow \mathcal{Q}(h')$.

Lemma 0.2. *Let $d \in \{0, 1, 2, \ldots, \infty\}$. The property of morphisms of germs*

$$\mathcal{P}_d((X, x) \to (S, s)) = \text{the local ring } \mathcal{O}_{X_s, x} \text{ of the fibre has dimension } d$$

is étale local on the source-and-target.

Proof. Given a diagram as in Definition 0.1 we obtain an étale morphism of fibres $U'_{v'} \to U_v$ mapping u' to u, see Lemma 04NI. Hence the result follows from Lemma 04N8. □